INTERNSHIP PROJECT

Enterprise Data Reconciliation

Turning messy transaction logs into clear financial insights

Python • Pandas • Regex • SQLite • Matplotlib

Presented by: Gummadavelli Srija

Internship submission

Problem Statement

- Enterprise transaction data is often large, unstructured, and inconsistent, making accurate financial reconciliation difficult.

Key challenges

1. Large Dataset
   - Processing around 100,000 records efficiently requires optimized data operations.
2. Unstructured Logs
   - Transaction details must be extracted from messy server logs using Regex.
3. Missing Weekend Exchange Rates
   - Weekend transactions need appropriate exchange-rate matching to avoid losing records.
4. Historical Customer Records
   - The latest customer record must be identified before matching transactions.

Goal: Build a clean, accurate, and unified dataset for financial analysis.

Project Solution

- Python and Pandas: Process and clean large datasets efficiently.
- Regex: Extract transaction details from unstructured server logs.
- SQL Window Functions: Identify the latest customer records and filter active users.
- Exchange-rate matching: Convert EUR transactions into USD and handle weekend dates appropriately.
- Matplotlib / Seaborn: Visualize monthly revenue and the top five customers.

Expected outcome: A reconciled financial dataset and clear revenue insights.

What does the project do?

The goal is to combine messy business data and turn it into useful financial information.

MAIN OBJECTIVE

Create one reliable view of transactions, customers, and USD revenue.

The source data includes raw server logs, daily exchange rates, and customer history.

TOOLS USED

Python • Pandas • Regex • SQLite • Matplotlib / Seaborn

WORKFLOW

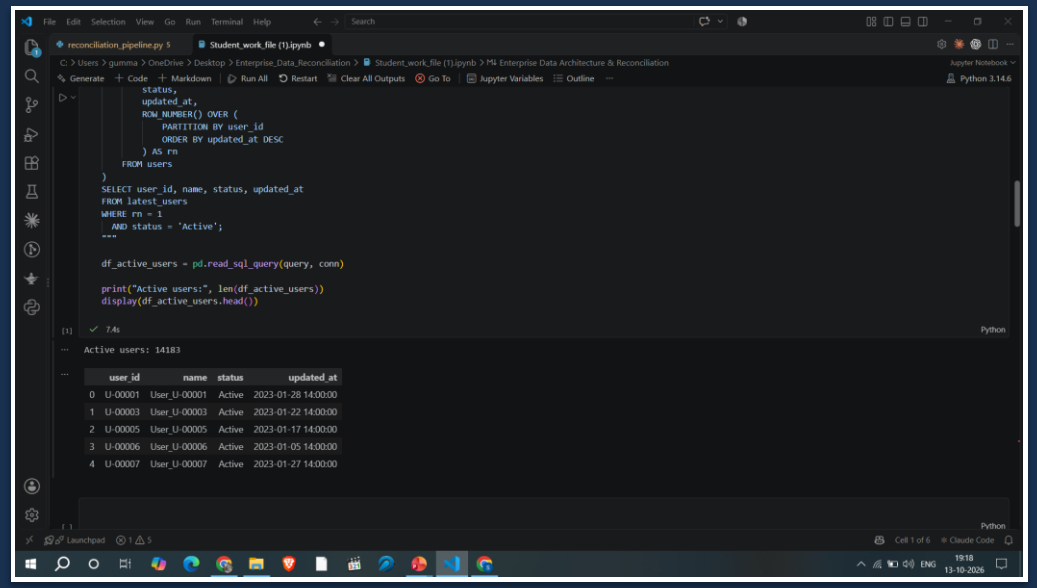
1. **Find latest customers**
   SQL window function
2. **Read and clean logs**
   Regex + Pandas
3. **Convert and reconcile**
   Exchange rates + joins
4. **Summarize results**
   Charts + tables

Result: a monthly revenue view and a top-customer summary.

Cleaning and extracting the source data

Screenshots from the notebook show the SQL and Regex steps.

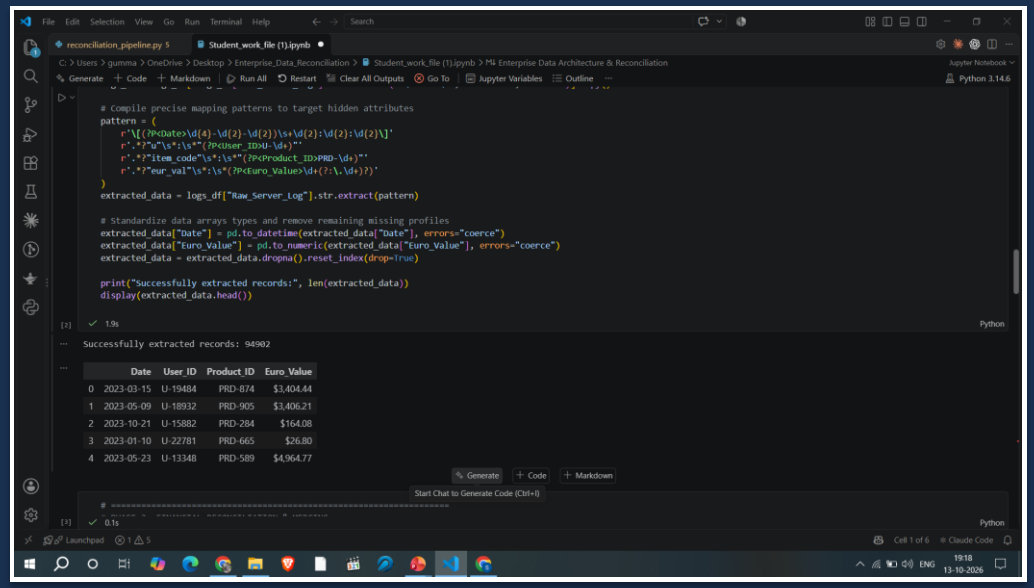
PHASE 1 • SQL CUSTOMER HISTORY



Selected the latest record per user and kept current Active users.

14,183 active users shown in output

PHASE 2 • REGEX LOG EXTRACTION

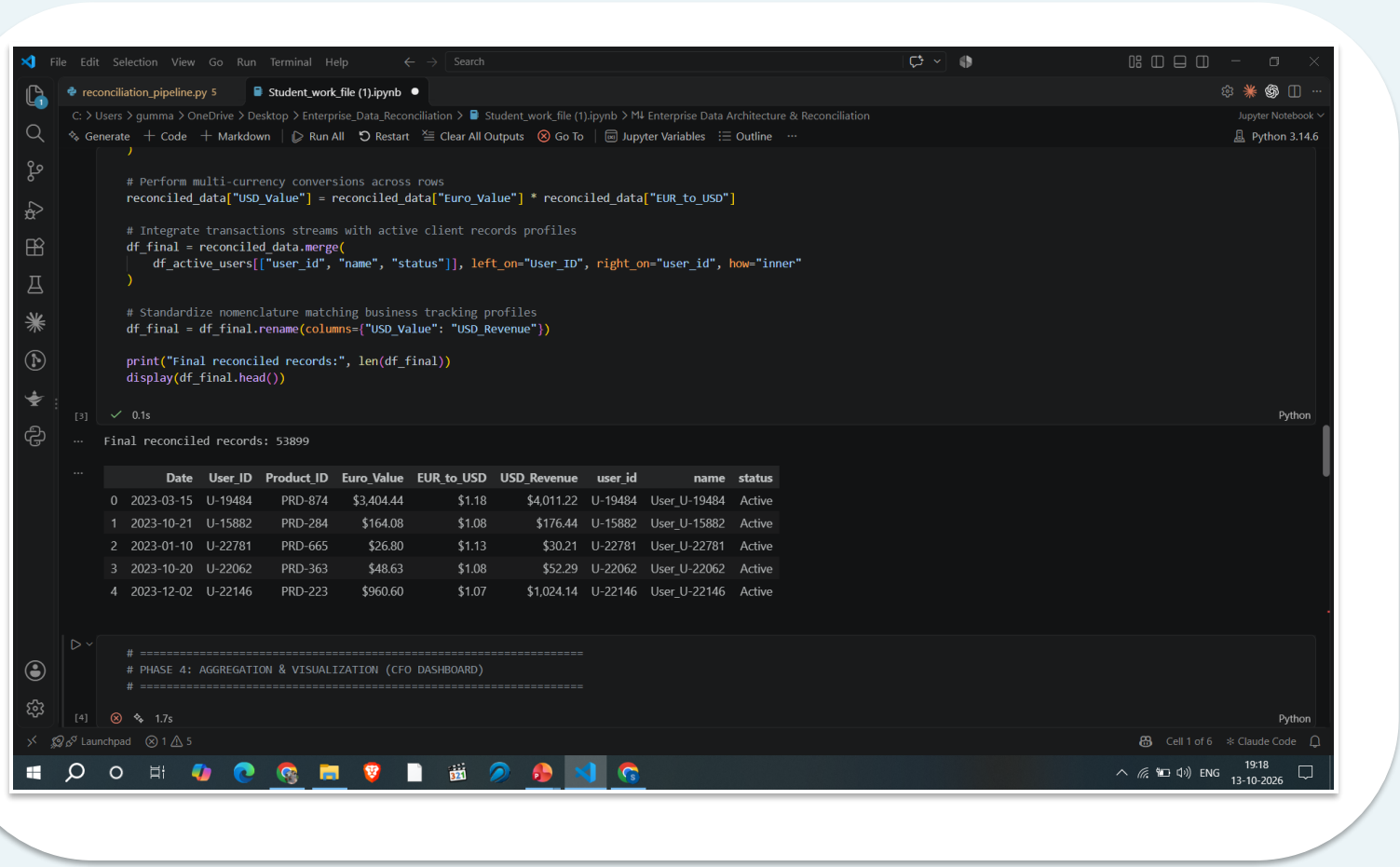


Extracted date, user, product, and euro amount from raw logs.

94,902 records extracted

Combining transactions, rates, and customers

The final table shows transaction values after currency conversion and customer matching.



KEY OUTPUT

53,899

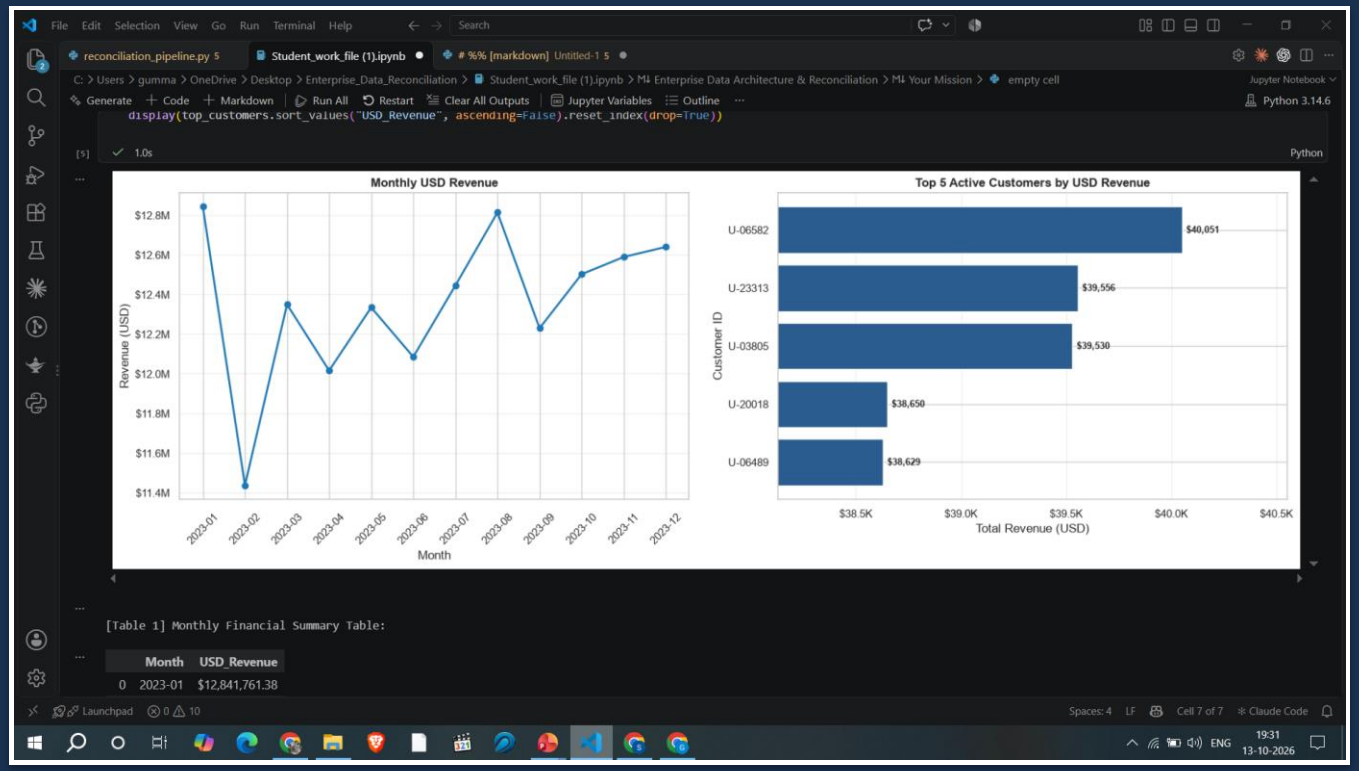
final reconciled records

EUR values were converted to USD and matched with active customer records.

Note: confirm weekend exchange-rate matching in the final run.

CFO dashboard: clear financial insights

The dashboard summarizes monthly revenue and highlights the top five customers.



Monthly revenue chart + top-five customer chart

KEY TAKEAWAYS

- Automated data extraction
- Combined transaction and customer data
- Created revenue summaries
- Visualized trends for review

Thank you

Enterprise Data Reconciliation

Internship Project Report

Prepared by: Gummadavelli Srija
Internship Submission

Python • Pandas • Regex • SQLite • Matplotlib/Seaborn

1. Abstract

This project focuses on turning raw and inconsistent business transaction data into a clearer financial view. It combines server logs, daily exchange rates, and customer history; extracts transaction details; converts EUR values to USD; matches transactions with active customer records; and summarizes the results through tables and charts.

2. Problem Statement

Business transaction data may be spread across raw server logs, exchange-rate files, and customer-history records. The data needs to be cleaned, structured, matched, and converted into a common currency before it can support reliable revenue analysis.

3. Objectives

- Identify the latest customer record and retain active customers.
- Extract transaction fields from raw logs using regular expressions and Pandas.
- Combine transaction, exchange-rate, and customer data.
- Convert euro transaction amounts to USD.
- Produce a reconciled dataset and visual summaries of monthly revenue and top customers.

4. Tools and Technologies

Python was used for data processing. Pandas supported data cleaning and transformation. Regular expressions (Regex) helped extract fields from raw logs. SQL window functions were used to select the latest customer record. SQLite supported querying, while Matplotlib/Seaborn were used for visual summaries.

5. Methodology

Phase 1 — Customer history: use a SQL window function to select the latest record for each user and keep current Active users.

Phase 2 — Log extraction: use Regex and Pandas to extract the date, user, product, and euro amount from raw server logs.

Phase 3 — Financial reconciliation: join transaction records with exchange rates and active customer records, then convert EUR values to USD.

Phase 4 — Reporting: summarize the reconciled data in monthly revenue and top-five customer charts.

6. Results

The presentation reports 14,183 active users after customer-history processing, 94,902 extracted log records, and 53,899 final reconciled records. The final dashboard is intended to show monthly revenue and the top five customers.

7. Conclusion

The project demonstrates a workflow for extracting, cleaning, joining, and reconciling business data, then presenting the results in a form that is easier to review. The process reduces the need to inspect raw logs manually and provides a consolidated view for financial analysis.

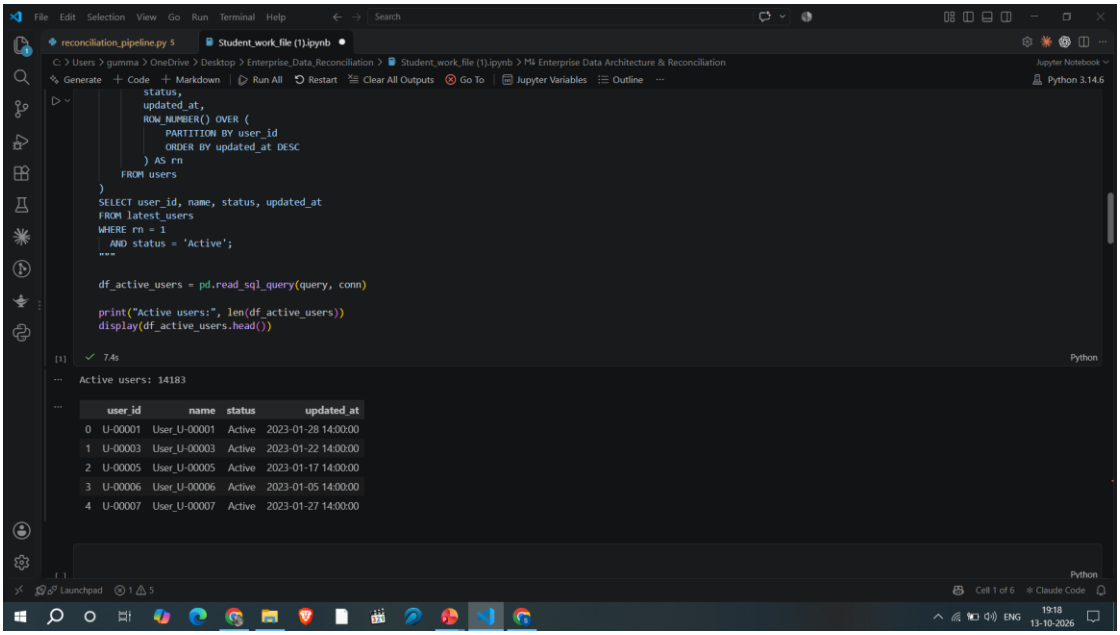
8. Validation Note

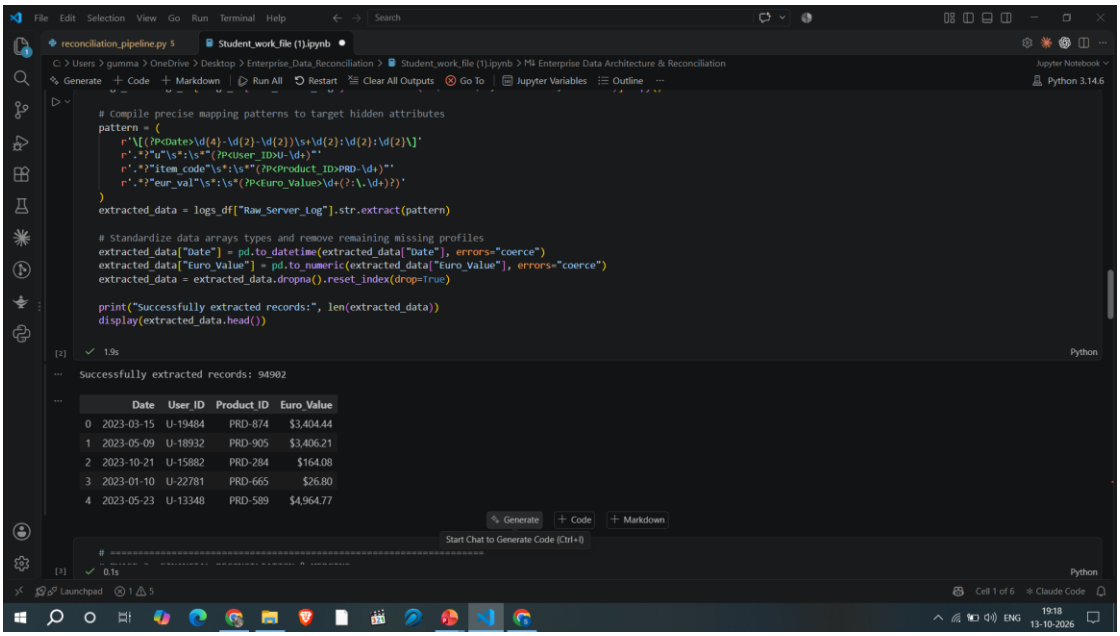
Before final submission, confirm how weekend dates are matched to exchange rates in the final notebook run. The presentation itself flags this as a point to verify. Only remove this note after the matching logic and final output have been checked.

9. Project Screenshots

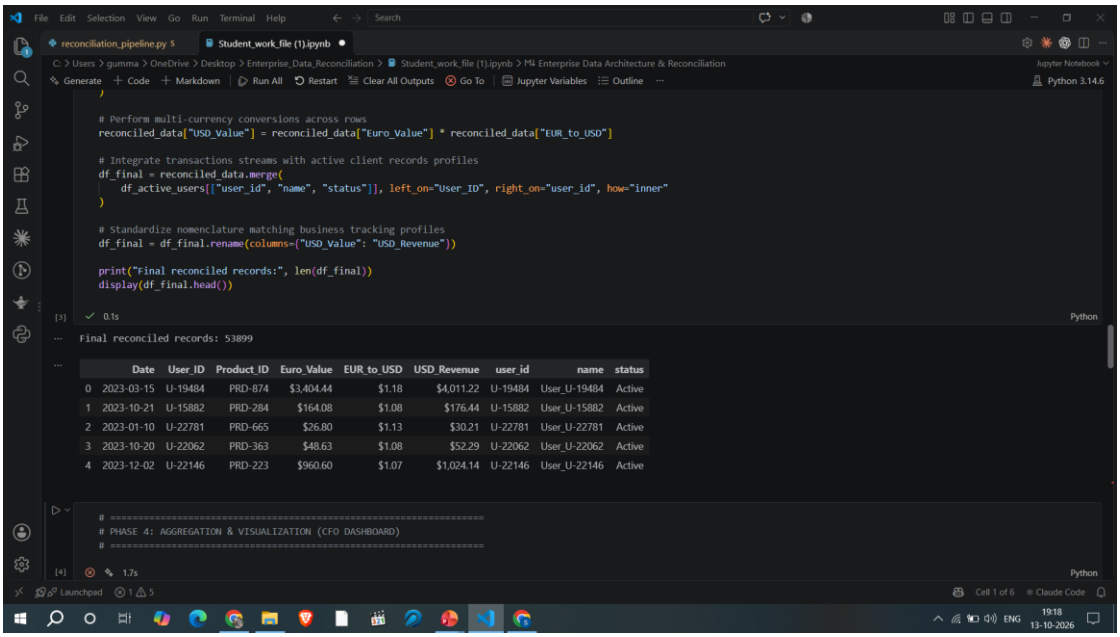
The following images are included from the uploaded presentation.

Slide 3 evidence





Slide 4 evidence



```
# Perform multi-currency conversions across rows
reconciled_data["USD_Value"] = reconciled_data["Euro_Value"] * reconciled_data["EUR_to_USD"]

# Integrate transactions streams with active client records profiles
df_final = reconciled_data.merge(
    df_active_users[["user_id", "name", "status"]], left_on="User_ID", right_on="user_id", how="inner"
)

# Standardize nomenclature matching business tracking profiles
df_final = df_final.rename(columns={"USD_Value": "USD_Revenue"})

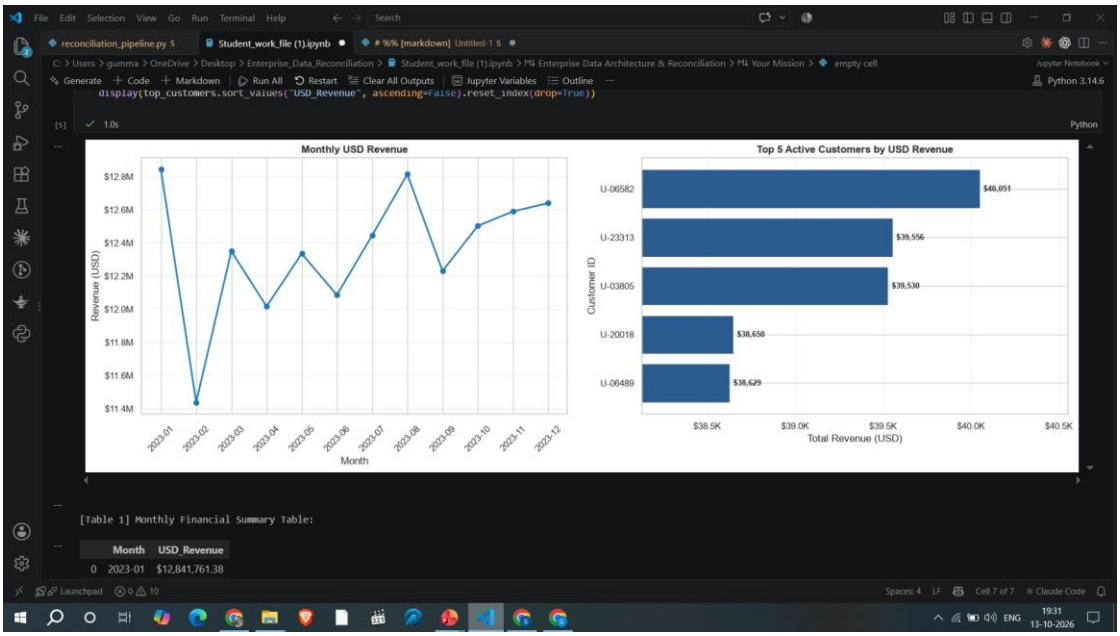
print("Final reconciled records:", len(df_final))
display(df_final.head())
```

```
Final reconciled records: 53899
```

	Date	User_ID	Product_ID	Euro_Value	EUR_to_USD	USD_Revenue	user_id	name	status
0	2023-03-15	U-19484	PRD-874	$3,404.44	$1.18	$4,011.22	U-19484	User_U-19484	Active
1	2023-10-21	U-15882	PRD-284	$164.08	$1.08	$176.44	U-15882	User_U-15882	Active
2	2023-01-10	U-22781	PRD-665	$26.80	$1.13	$30.21	U-22781	User_U-22781	Active
3	2023-10-20	U-22062	PRD-363	$48.63	$1.08	$52.29	U-22062	User_U-22062	Active
4	2023-12-02	U-22146	PRD-223	$960.60	$1.07	$1,024.14	U-22146	User_U-22146	Active

```
# ============================================================
# PHASE 4: AGGREGATION & VISUALIZATION (CFO DASHBOARD)
# ============================================================
```

Slide 5 evidence




```
display(top_customers.sort_values("USD_Revenue", ascending=False).reset_index(drop=True))
```

```
[Table 1] Monthly Financial Summary Table:
```

	Month	USD_Revenue
0	2023-01	$12,841,761.38